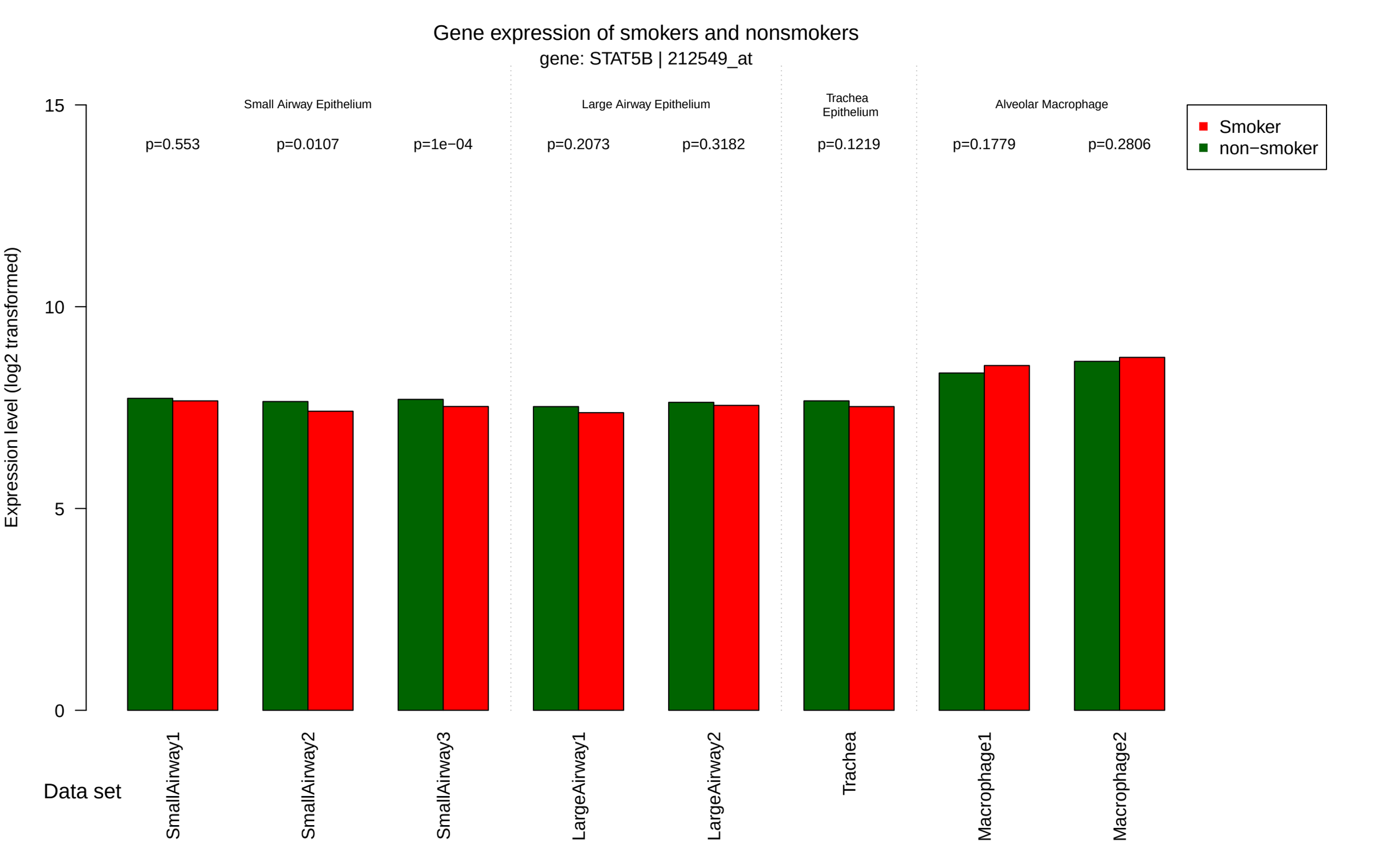
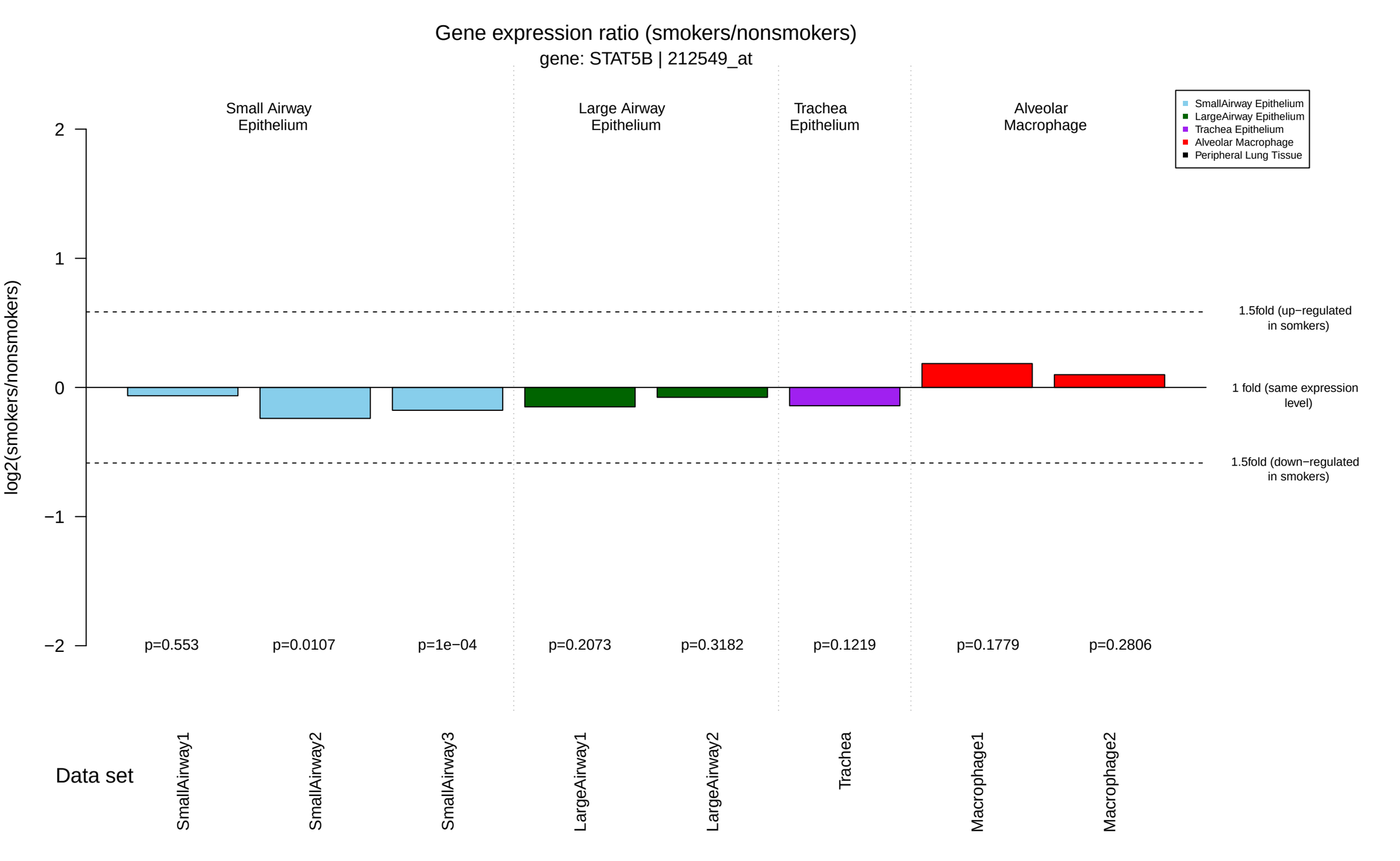
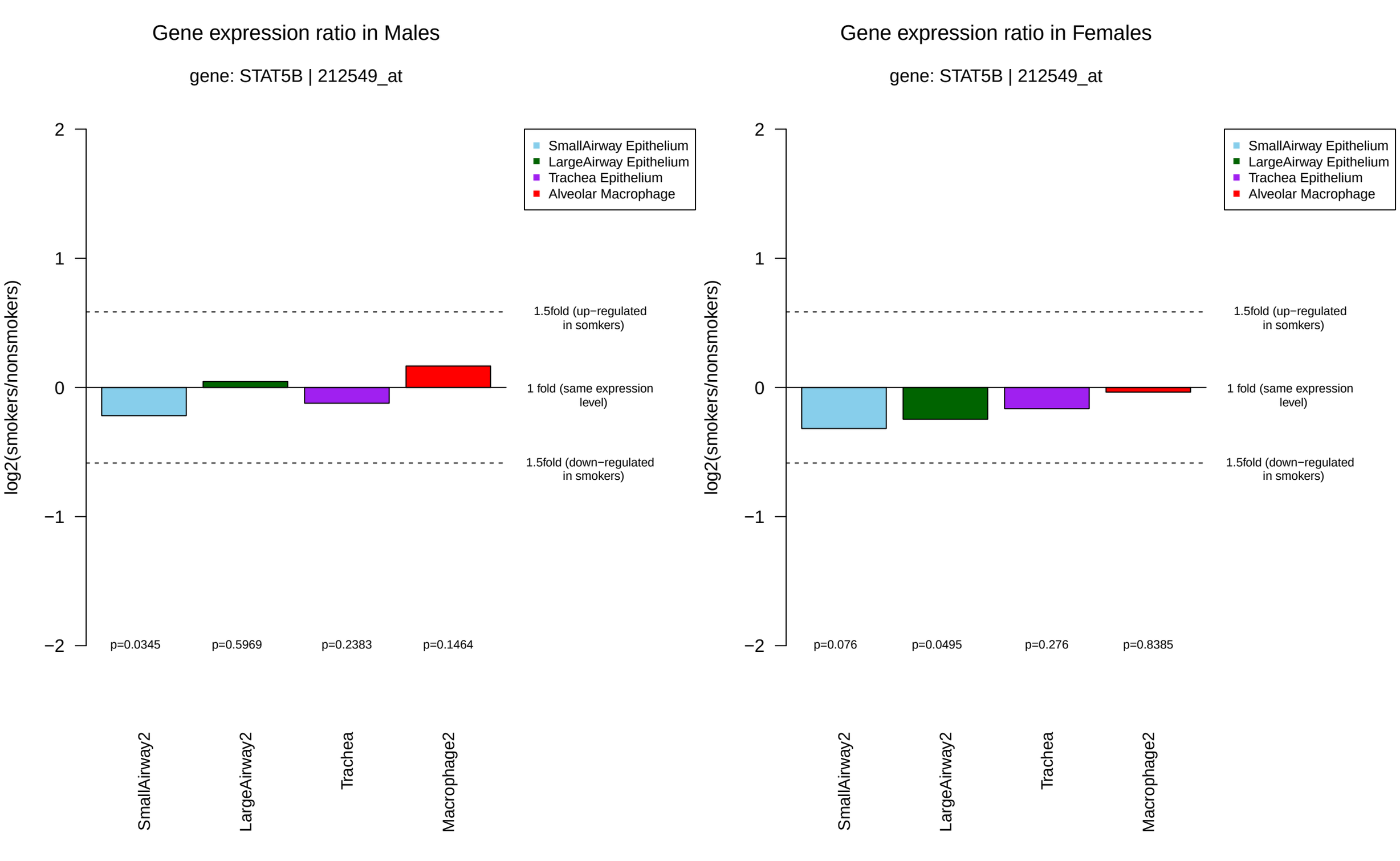
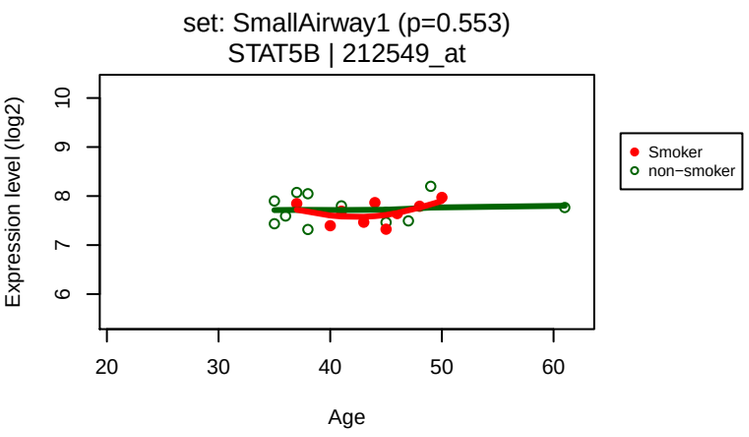
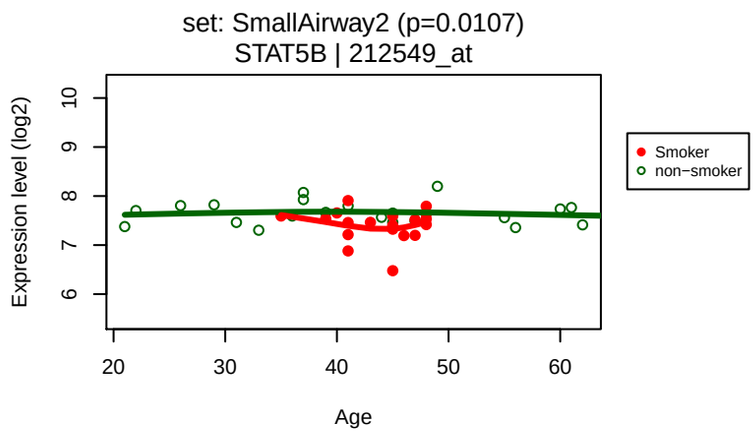
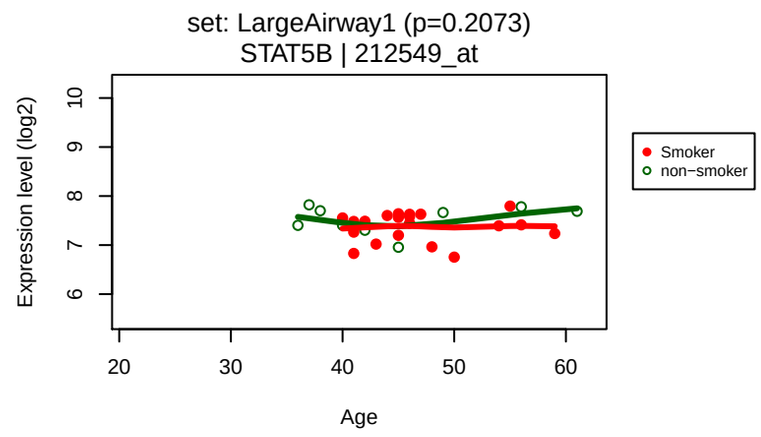

Gene expression of smokers and nonsmokers
gene: STAT5B | 212549_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.553
p=0.0107
p=1e−04
p=0.2073
p=0.3182
p=0.1219
p=0.1779
p=0.2806
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

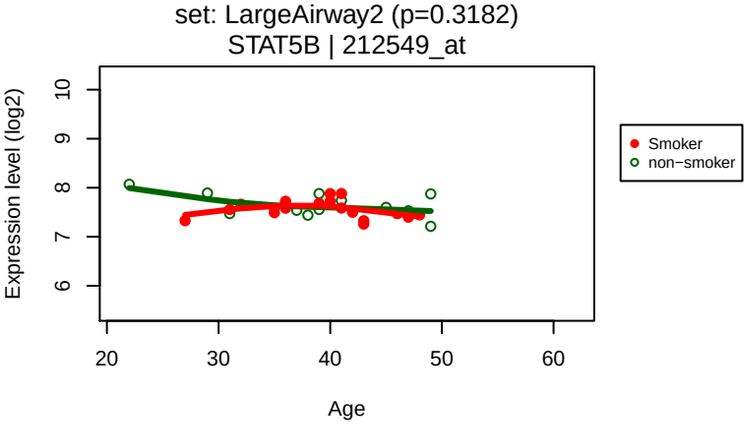

Gene expression ratio (smokers/nonsmokers)
gene: STAT5B | 212549_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.553
p=0.0107
p=1e−04
p=0.2073
p=0.3182
p=0.1219
p=0.1779
p=0.2806
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

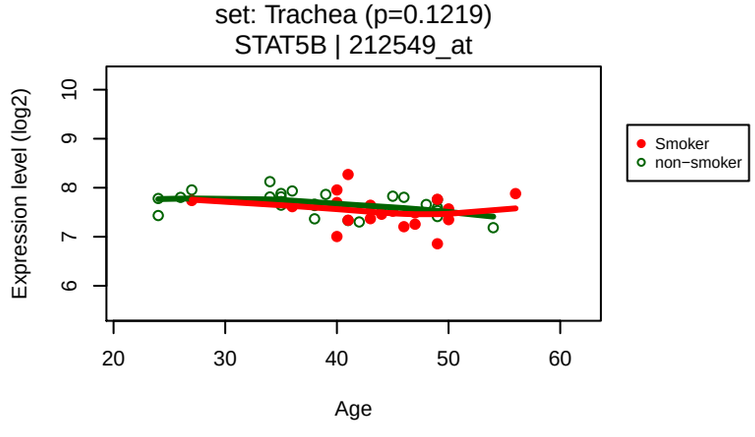

Gene expression ratio in Males
gene: STAT5B | 212549_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0345
p=0.5969
p=0.2383
p=0.1464
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT5B | 212549_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.076
p=0.0495
p=0.276
p=0.8385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

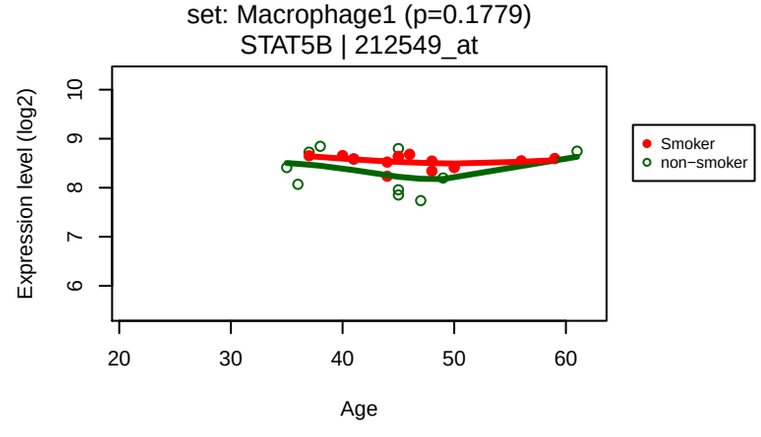

set: SmallAirway1 (p=0.553)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker

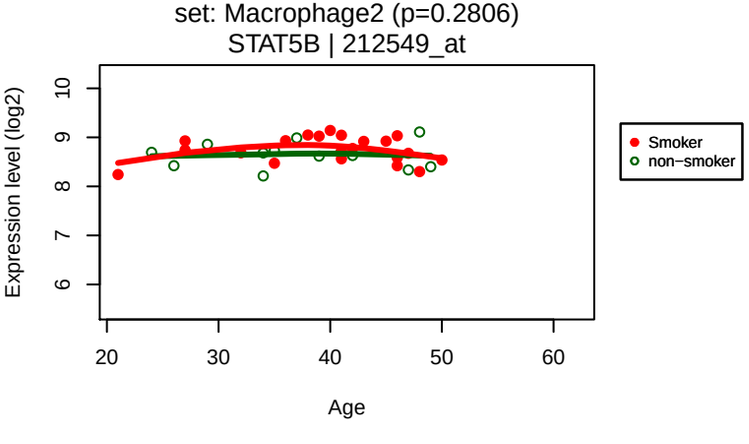

set: SmallAirway2 (p=0.0107)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker

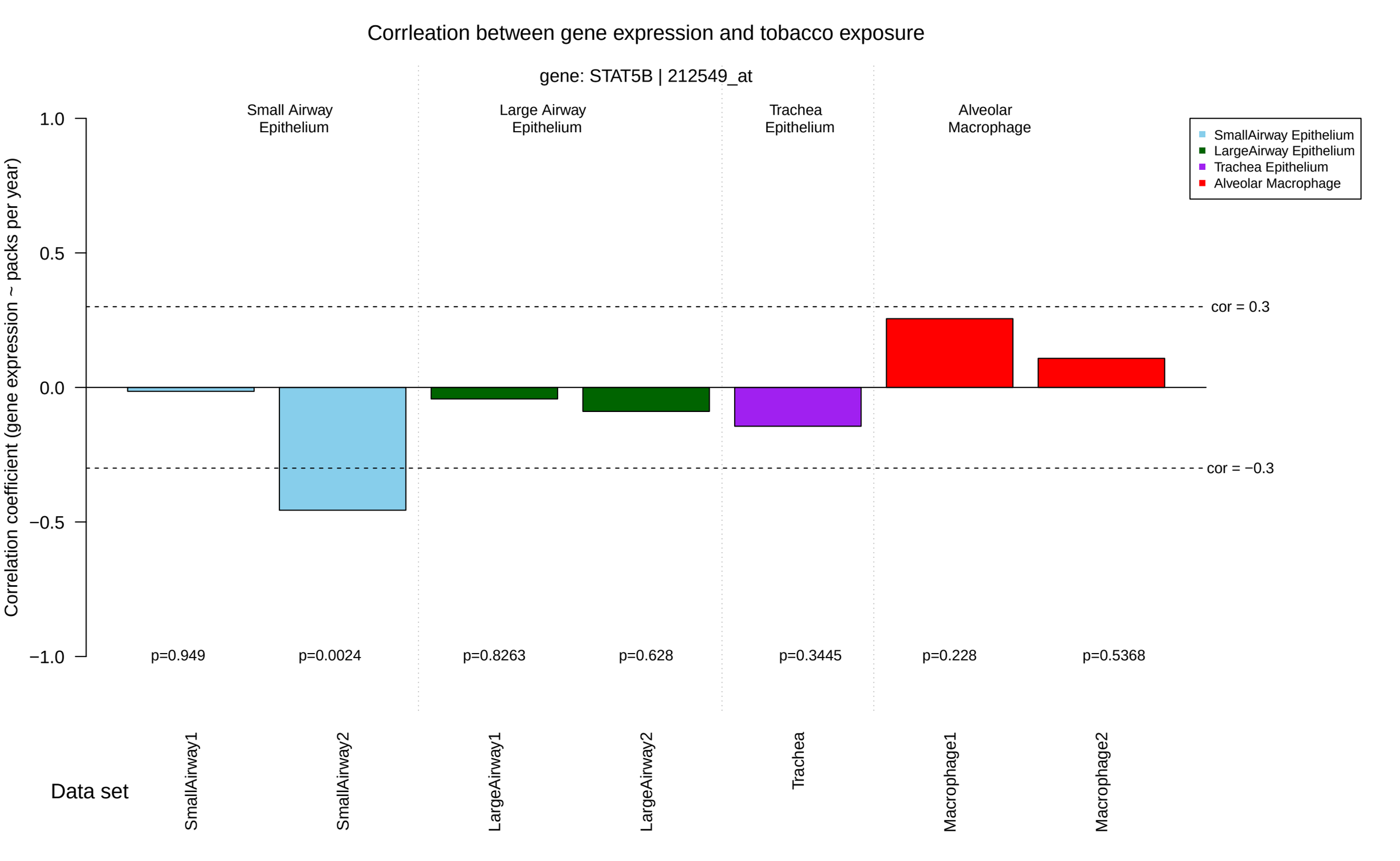

set: LargeAirway1 (p=0.2073)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3182)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1219)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1779)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2806)
STAT5B | 212549_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT5B | 212549_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.949
p=0.0024
p=0.8263
p=0.628
p=0.3445
p=0.228
p=0.5368
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2